

EDUCATION

University of British Columbia (UBC)

MSc in Computer Science with research awards for the entire program

Graduation: May 2025

GPA: 91%

McGill University

BSc in Joint Major of Computer Science and Statistics

Graduation: May 2022

GPA: 3.96/4.0

Awards: Dean's Honor List for top 10% graduates in the Faculty of Science, NSERC Undergraduate Research Award, Emily Ross Crawford Scholarship for high academic merit, Tomlinson Engagement Award for tutoring

WORK EXPERIENCE

Graduate Researcher

UBC - Software Practices Lab

Sept 2022 - Mar 2025

- Published 3 fuzzing and program analysis papers in top-tier venues (TOSEM, ISSTA, <30% acceptance)
- Enhanced a Docker-based testing framework by integrating recent fuzzers for input generation, execution monitoring, and anomaly detection, improving test efficiency and reducing resource waste by 71%
- Optimized search algorithms for execution trace analysis, reducing runtime from 20 min to 2 min
- Created a novel metric to quantify mutation destructiveness in generator-based fuzzing, revealing trade-offs between early program exploration and deep vulnerability discovery in open source projects
- Contributed to issues in [Magma](#) (300+ GitHub stars) to improve stability, maintainability, and performance

Software Developer

Morgan Stanley - Montreal, Canada

May 2021 - Dec 2021

- Reduced inspection time by building monitoring dashboards for anomaly detection in 1M⁺ daily trades
- Developed a terminal module in an Angular web application for database host provisioning
- Improved code coverage by ~20% by implementing Jenkins CI integration testing with automated reports
- Top 5 in intern AI contest; built strategic agent for Ultimate Tic-Tac-Toe using search algorithms

Software Developer

Citibank - Mississauga, Canada

Jul 2020 - Aug 2020

- Designed a web application with AWS chatbot, providing personalized investment advice for remote clients
- Processed 9-year time series stock data with Python and visualized portfolio with Power BI dashboards

TEACHING EXPERIENCE AND PROJECTS

- **Teaching Assistantship (2020 - 2024):** Collaborated with faculty and teams of 4 - 34 TAs to mentor software projects, review code, and guide system design to help students understand complex concepts.
 - Assisted Courses: CPSC410 Advanced Software Engineering (UBC, 2024), CPSC210 Software Construction (UBC, 2022), COMP303 Software Design (McGill, 2020).
- **WWTP Dashboards (2022):** Developed a scalable React-based dashboard visualizing global wastewater treatment data; integrated Python-processed metrics for dynamic, country-level comparisons. React, Node.js
- **JetUML Desktop App (2020):** Implemented layout algorithms for sequence diagrams in a software in software design courses at McGill, improving usability and rendering accuracy. Java, JavaFX

TECH SKILLS

Languages: Python, Java, C++, JavaScript

Technologies: Git, Docker, Jenkins, JUnit, Pandas, NumPy, React, Node.js, REST APIs, HTML

Relevant Courses: Data Structures, Software Design, Algorithm Design, Computer Systems, Statistics

EXTRACURRICULARS

- Facilitator in the software engineering reading group at the PLSE lab in the University of Washington (2024)
- Vice President of Sponsorship of Hack4Impact at McGill, which builds software for nonprofits (2021)